

Data analysis Project

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1 What is the aim of this project ?

First, let us recall the aim of this project : we want to **statistically investigate** the following question :

What aspects of daily life can significantly influence students' academic performance ?

To answer this question, we will develop a statistical methodology to analyze several factors that may be associated with academic performance. The following sections present the study framework and the statistical methods used.

2 Definition of the Study Framework

This study is based on a sample of 78 I1 students who answered an anonymous survey about their daily habits and academic performance. Among the 15 variables collected, we focus our analysis on the following key metrics :

Variable	Type	Role
Average grade of last semester	Quantitative	DV
Time spent in extracurricular activities per week	Quantitative	IV
Transportation time from home to Efrei	Quantitative	IV
Average sleeping time (outside exam period)	Quantitative	IV
Average sleeping time during exam period	Quantitative	IV

Table 1: Variables used among those collected through the survey

Legend:

- Dependent Variable (DV): variable to be explained;
- Independent Variable (IV): explanatory variable used to explain the DV.

2.1 Potential Biases

Potential biases include:

- Selection bias due to the limited sample size;
- Social desirability bias, as students may provide answers considered more socially acceptable;
- Declaration bias caused by inaccurate self-evaluation or estimation;
- Limited representativeness of the sample population.

3 Statistical methodology

The objective of this study is to determine whether some lifestyle variables are statistically associated with students' academic performance. Formulating daily life questions into statistical terms is essential to transition from subjective intuition to objective scientific inquiry. By defining mathematical parameters (such as a regression slope or a correlation coefficient), we can quantify relationships, rigorously account for sampling fluctuations, and apply universal decision rules (using risks and p-values) to accept or reject hypotheses.

- T_1 : extracurricular activities;
- T_2 : transportation time;
- S_1 : sleep duration outside exam periods;
- S_2 : sleep duration during exam periods;
- G : average grade.

Academic behavior variables (study time, attendance, etc.) are excluded to avoid mixing causes (study effort) with external factors (lifestyle conditions).

For this study, qualitative data may be changed into quantitative data, while some quantitative data may also be changed into categorical data for chi-square tests. The transformation of variables may lead to information loss and reduced statistical precision.

We first study the relationship between academic performance and each quantitative lifestyle variable separately using simple linear regression models of the form:

$$G = a + bX + \varepsilon$$

where:

- G is the average grade;
- X is one explanatory variable;
- ε is the error term.

3.1 Statistical Hypotheses

For each variable X , we test $H_0 : b = 0$ and $H_1 : b \neq 0$ where:

- H_0 means that there is no linear relationship between X and G ;
- H_1 means that there is a significant linear relationship between X and G .

The decision is made using a 95% confidence interval. The null hypothesis is rejected if 0 does not belong to the confidence interval.

$$IC_{95\%}(b) = \hat{b} \pm t_{n-2, 0.975} \times SE(\hat{b})$$

where $SE(\hat{b})$ is the standard error such that :

$$SE(\hat{b}) = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{(n-2) \sum_{i=1}^n (x_i - \bar{x})^2}}$$

3.2 Inference Tools

For each regression, we use:

- Simple linear regressions;
- 95% confidence intervals for coefficient estimation.

We also perform a multiple linear regression model including all quantitative explanatory variables simultaneously:

$$G = a + b_1T_1 + b_2T_2 + b_3S_1 + b_4S_2 + \varepsilon$$

The validity of the models relies on assumptions of independence, linearity, and homoscedasticity. Furthermore, since our sample size is large enough ($n = 78 \geq 30$), the Central Limit Theorem applies, allowing us to use hypothesis testing.

3.3 Chi-Square Tests

To complement the regression analysis, some quantitative variables were transformed into categorical variables in order to perform chi-square tests of independence.

We test the independence between:

- sleep level and academic performance;
- transportation time and academic performance;
- extracurricular activity level and academic performance.

For each chi-square test:

H_0 : the two variables are independent

H_1 : there is an association between the variables

Decision rule and p-value interpretation:

The chi-square test is based on a test statistic that follows a chi-square distribution under H_0 . Instead of comparing the test statistic to a critical value from tables, we can also use the **p-value** approach.

The p-value represents the probability of observing a test statistic at least as extreme as the one computed, assuming that H_0 is true.

Decision rule:

If $p \leq 0.05$, reject H_0 ; otherwise, do not reject H_0 .

The chi-square test of independence and p-values are used for statistical decision-making. To ensure the validity of the Chi-square tests, we verified that the necessary theoretical conditions were met: the total sample size is sufficient ($n = 78 \geq 50$), and we ensured that all expected frequencies in our contingency tables were greater than or equal to 5 ($n_{i,j} \geq 5$), regrouping categories when strictly necessary.

3.4 Software and tools used

The statistical analysis was performed using Python with the following libraries:

- pandas
- numpy
- scipy

4 Results and Interpretation

4.1 Descriptive Statistics

Before conducting inferential tests, we summarize the main quantitative variables of our sample ($n = 78$). The average grade (G) is roughly 14.22/20 ($SD = 1.83$). Students sleep on average 6.71 hours outside exam periods (S_1) and 5.95 hours during exams (S_2). Extracurricular activities (T_1) take around 3.63 hours/week, and transportation (T_2) averages 0.74 hours.

4.2 Linear regression: Grade and sleep duration (S1)

The linear regression between grades and average sleeping time outside exam periods gives:

- $R^2 = 0.0412$
- 95% confidence interval for the slope: $IC = [-0.038; 0.781]$
- p-value = 0.075

This result is not statistically significant at the 5% level, suggesting a possible weak relationship.

4.3 Linear regression: Grade and sleep during exam period (S2)

The regression between grades and sleeping time during exam periods gives:

- $R^2 = 0.0479$
- 95% confidence interval for the slope: $IC = [-0.006; 0.666]$
- p-value = 0.054

This result is not statistically significant at the 5% level, but is very close to the threshold, suggesting a possible weak relationship.

4.4 Linear regression: Grade and extracurricular activities (T1)

The regression between grades and time spent in extracurricular activities per week gives:

- $R^2 = 0.0143$
- 95% confidence interval for the slope: $IC = [-0.089; 0.288]$
- p-value = 0.297

No statistically significant relationship is found between extracurricular activities and academic performance.

4.5 Linear regression: Grade and transportation time (T2)

The regression between grades and transportation time gives:

- $R^2 = 0.0389$
- 95% confidence interval for the slope: $IC = [-1.455; 0.092]$
- p-value = 0.083

No statistically significant relationship is observed at the 5% level.

4.6 Multiple linear regression (OLS)

The multiple linear regression model estimated using Ordinary Least Squares (OLS) is:

$$G = 12.1833 + 0.0537T_1 - 0.4868T_2 + 0.2064S_1 + 0.1369S_2$$

4.7 Chi-square tests of independence

To complement the regression analysis, chi-square tests of independence were performed on categorical variables.

Hypotheses:

H_0 : the variables are independent

H_1 : the variables are dependent (associated)

The null hypothesis is rejected when $p \leq 0.05$.

- G and S_1 : $\chi^2 = 18.708$, p-value = 0.096 $\rightarrow$ not significant
- G and S_2 : $\chi^2 = 20.968$, p-value = 0.051 $\rightarrow$ not significant (borderline)
- G and T_1 : $\chi^2 = 9.714$, p-value = 0.641 $\rightarrow$ not significant
- G and T_2 : $\chi^2 = 10.220$, p-value = 0.597 $\rightarrow$ not significant

5 Discussion and Conclusion

Overall, the results suggest that there is no strong statistically significant relationship between students' academic performance and the lifestyle variables studied in this project. Although some variables such as sleep duration (especially during exam periods) show weak positive trends, these effects remain statistically insignificant at the 5% level. The coefficient associated with transportation time is negative, suggesting that longer commuting times may be associated with lower academic performance. However, this effect is not statistically significant.

The low R^2 values obtained in all simple linear regressions indicate that the explanatory power of the models is very limited. This means that the studied variables explain only a small proportion of the variation in students' grades.

Similarly, the chi-square tests of independence do not provide sufficient evidence of association between categorical versions of the variables and academic performance. Even though the relationship between sleep during exam periods and grades is close to significance, it is still not statistically conclusive.

Finally, the multiple linear regression model shows intuitive signs (positive effect of sleep, negative effect of transportation time), but these coefficients should be interpreted with caution due to the lack of statistical significance and possible correlations between explanatory variables.

6 Appendix

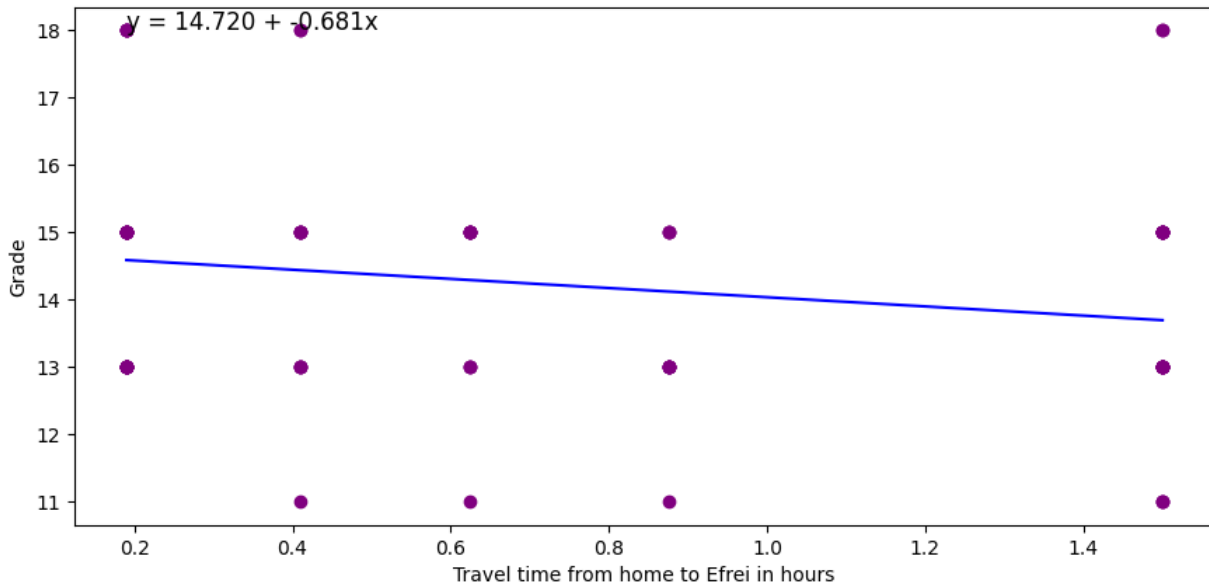


Figure 1: Linear regression between average grade and traveling time in hours

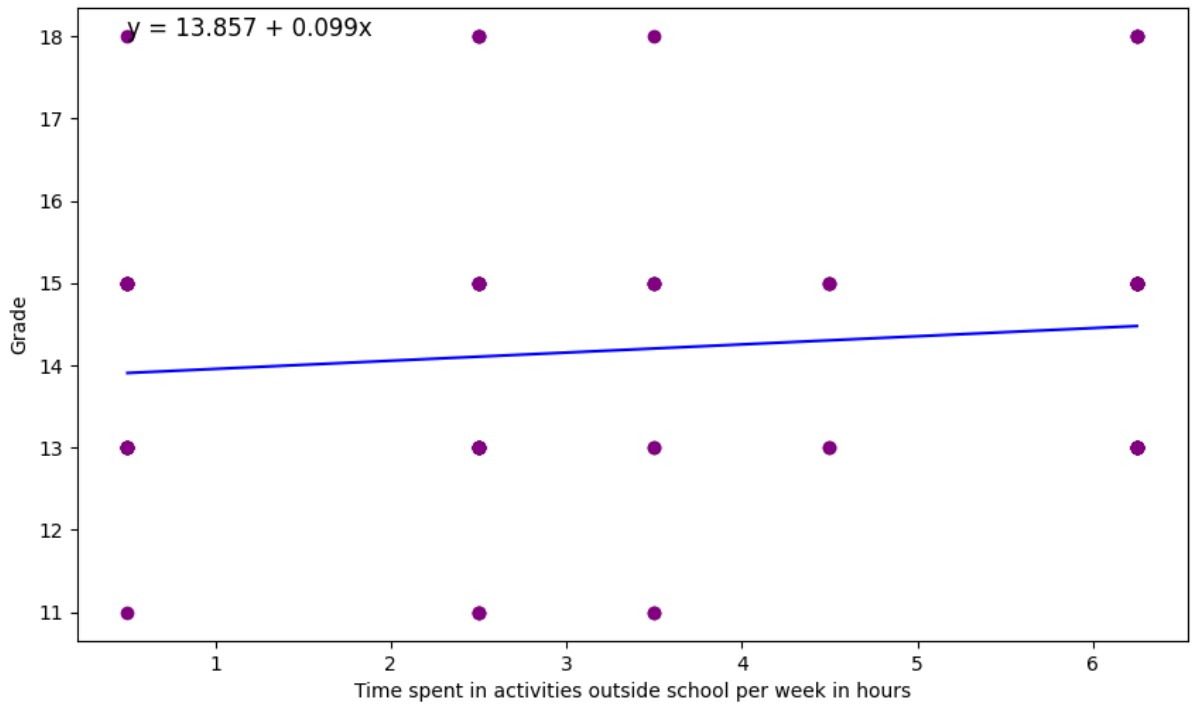


Figure 2: Linear regression between average grade and extracurricular activity time in hours

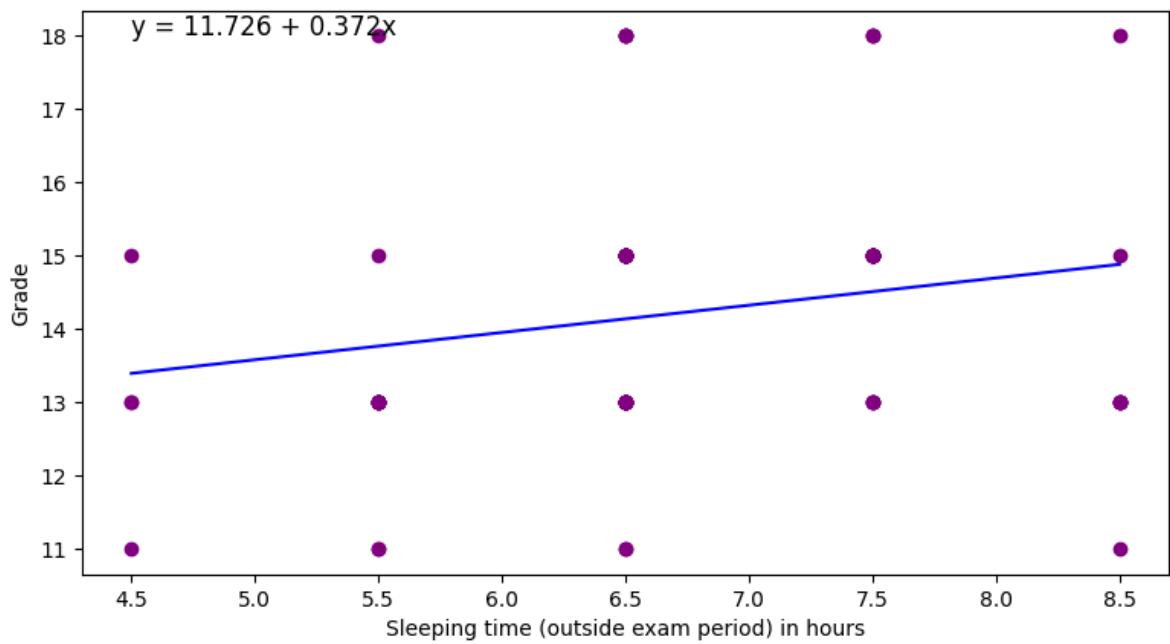


Figure 3: Linear regression between average grade and sleeping time outside exam periods in hours

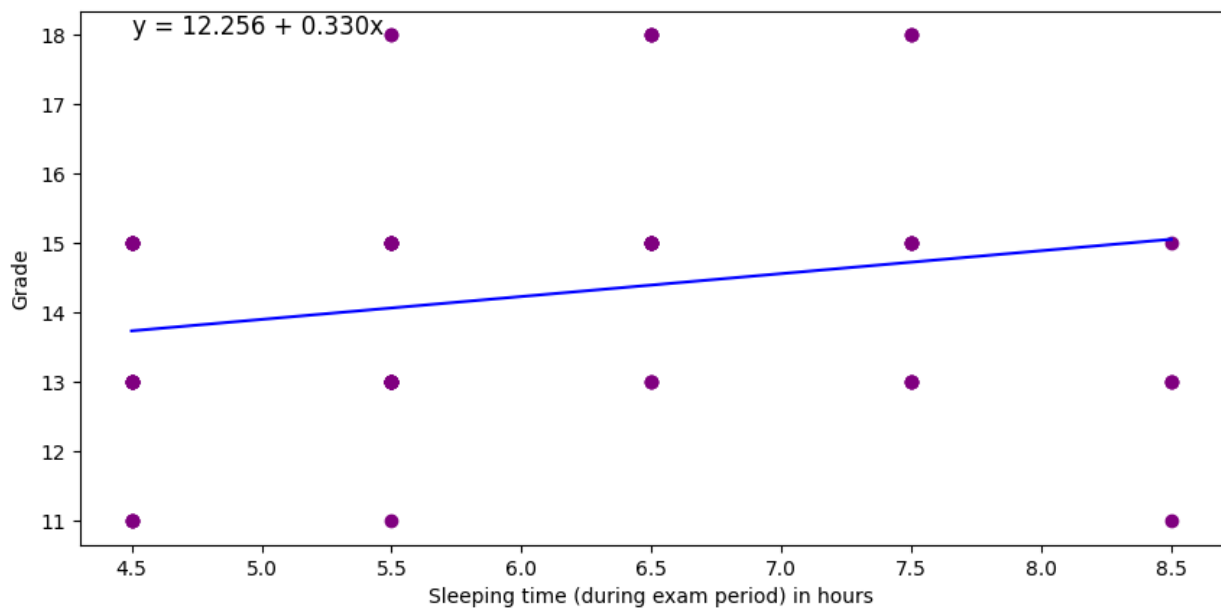


Figure 4: Linear regression between average grade and sleeping time during exam periods in hours

```

Coefficients :
const          12.183273
timeActivity    0.053723
transportationTime -0.486822
sleepingTime    0.206385
sleepingTimeExams 0.136920
dtype: float64

R² : 0.0783156321193279

```

Figure 5: Coefficients of the OLS

```
(envi) jalil@daPC:~/pythondev/data$ /home/jalil/pythondev/data/envi/bin/python /home/jalil/pythondev/data/data.py
OLS Regression Results

=====
Dep. Variable:          grade    R-squared:          0.078
Model:                  OLS      Adj. R-squared:       0.028
Method:                 Least Squares    F-statistic:        1.551
Date:                   Mon, 11 May 2026    Prob (F-statistic):  0.197
Time:                   09:27:25    Log-Likelihood:     -154.04
No. Observations:       78    AIC:                  318.1
Df Residuals:           73    BIC:                  329.9
Df Model:                4
Covariance Type:        nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	12.1833	1.531	7.958	0.000	9.132	15.234
timeActivity	0.0537	0.096	0.561	0.577	-0.137	0.245
transportationTime	-0.4868	0.410	-1.189	0.238	-1.303	0.329
sleepingTime	0.2064	0.272	0.759	0.450	-0.335	0.748
sleepingTimeExams	0.1369	0.233	0.588	0.558	-0.327	0.601

```
=====
Omnibus:                 3.092    Durbin-Watson:       2.111
Prob(Omnibus):           0.213    Jarque-Bera (JB):     2.771
Skew:                    0.462    Prob(JB):              0.250
Kurtosis:                2.979    Cond. No.              75.0
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 6: Whole study of the OLS

7 Python code

```
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import chi2_contingency

# Loading dataset
df = pd.read_csv("./data.csv")

# Mapping categorical intervals into numerical values
gradeMap = {
    "Superieure a 16" : 18,
    "Entre 14 et 16" : 15,
    "Entre 12 et 14" : 13,
    "Entre 10 et 12" : 11,
}

timeActivityMap = {
    "5 heures ou plus" : 6.25,
    "4 a 5 heures" : 4.5,
    "3 a 4 heures" : 3.5,
    "2 a 3 heures" : 2.5,
    "1 heure ou moins" : 0.5,
}

transportationTimeMap = {
    "1 heure ou plus" : 1.5,
    "45 minutes a 1 heure" : 0.875,
    "30 a 45 minutes" : 0.625,
    "20 a 30 minutes" : 0.41,
    "5 a 20 minutes" : 0.19,
}

sleepingTimeMap = {
    "Plus de 8 heures" : 8.5,
```



```

    "Entre 7 et 8 heures" : 7.5,
    "Entre 6 et 7 heures" : 6.5,
    "Entre 5 et 6 heures" : 5.5,
    "Entre 4 et 5 heures" : 4.5
}

# Data conversion
df["grade"] = df["grade"].map(gradeMap)
df["timeActivity"] = df["timeActivity"].map(timeActivityMap)
df["transportationTime"] = df["transportationTime"].map(transportationTimeMap)
df["sleepingTime"] = df["sleepingTime"].map(sleepingTimeMap)
df["sleepingTimeExams"] = df["sleepingTimeExams"].map(sleepingTimeMap)

# Removing missing values
df = df.dropna()
# -----
# DESCRIPTIVE STATISTICS
# -----
print("=====")
print("          DESCRIPTIVE STATISTICS          ")
print("=====")

cols_to_describe = [
    "grade",
    "sleepingTime",
    "sleepingTimeExams",
    "timeActivity",
    "transportationTime"
]

for col in cols_to_describe:
    mean_val = df[col].mean()
    std_val = df[col].std()
    print(f"{col} : Moyenne = {mean_val:.2f} | Ecart-type (SD {std_val:.2f})")

print("=====\n")

# -----
# SIMPLE LINEAR REGRESSION
# -----

X = df["transportationTime"]
y = df["grade"]

X_const = sm.add_constant(X)

model = sm.OLS(y, X_const).fit()

print(model.summary())

# Rsquared
print("Rrsquared =", model.rsquared)

# Confidence interval
ic = model.conf_int().loc["transportationTime"]

print("95% confidence interval :", ic)

# -----
# PLOT
# -----

plt.scatter(X, y, color="purple")

X_sorted = np.sort(X)

```

```

X_sorted_const = sm.add_constant(X_sorted)

y_pred = model.predict(X_sorted_const)

plt.plot(X_sorted, y_pred, color="blue")

plt.xlabel("Travel time from home to Efrei in hours")
plt.ylabel("Grade")

plt.show()

# -----
# MULTIPLE OLS REGRESSION
# -----

X_multi = df[
    [
        "timeActivity",
        "transportationTime",
        "sleepingTime",
        "sleepingTimeExams"
    ]
]

X_multi = sm.add_constant(X_multi)

model_multi = sm.OLS(y, X_multi).fit()

print(model_multi.summary())

# -----
# CHI-SQUARE TEST OF INDEPENDENCE
# -----

# Creating categorical variables
df["gradeCategory"] = pd.cut(
    df["grade"],
    bins=[0, 12, 14, 20],
    labels=["Low", "Medium", "High"]
)

df["sleepCategory"] = pd.cut(
    df["sleepingTime"],
    bins=[0, 5, 7, 10],
    labels=["Low sleep", "Medium sleep", "High sleep"]
)

# Contingency table
contingency_table = pd.crosstab(
    df["gradeCategory"],
    df["sleepCategory"]
)

# Chi-square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

print("Chi-square statistic :", chi2)
print("p-value :", p)

if p <= 0.05:
    print("Reject H0 : variables are associated.")
else:
    print("Do not reject H0 : variables are independent.")

```

7.1 Mathematical Derivation of the Multiple OLS Estimator

In Section 3.2, we introduced the multiple linear regression model containing our four lifestyle explanatory variables:

$$G = a + b_1T_1 + b_2T_2 + b_3S_1 + b_4S_2 + \varepsilon \quad (1)$$

To compute the vector of optimal coefficients using Ordinary Least Squares (OLS), we express the system of $n = 78$ observations using matrix algebra:

$$Y = X\beta + \epsilon \quad (2)$$

Where:

- Y is a $(n \times 1)$ column vector containing the average grades (G).
- X is a $(n \times 5)$ design matrix where the first column is filled with 1s (to estimate the intercept a), and the subsequent columns contain the observations for T_1 , T_2 , S_1 , and S_2 .
- $\beta = [a, b_1, b_2, b_3, b_4]^T$ is the $((4 + 1) \times 1)$ vector of parameters to estimate.
- ϵ is the $(n \times 1)$ vector of residuals.

The OLS optimization framework aims to minimize the Residual Sum of Squares (RSS), defined as the inner product of the error vector:

$$\begin{aligned} RSS(\beta) &= \epsilon^T \epsilon = (Y - X\beta)^T (Y - X\beta) \\ &= Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X\beta \end{aligned} \quad (3)$$

Since the quantity $Y^T X\beta$ is a scalar (1×1), it is strictly equal to its transpose $(Y^T X\beta)^T = \beta^T X^T Y$. We can therefore simplify the objective function:

$$RSS(\beta) = Y^T Y - 2\beta^T X^T Y + \beta^T X^T X\beta \quad (4)$$

To find the estimator $\hat{\beta}$ that minimizes this sum of squared errors, we calculate the gradient with respect to β and set it to zero:

$$\frac{\partial RSS}{\partial \beta} = -2X^T Y + 2X^T X\hat{\beta} = 0 \quad (5)$$

This leads directly to the matrix form of the **Normal Equations**:

$$X^T X\hat{\beta} = X^T Y \quad (6)$$

Assuming that the matrix $X^T X$ has full column rank (meaning there is no perfect multicollinearity among our lifestyle variables), $X^T X$ is invertible. Multiplying both sides from the left by $(X^T X)^{-1}$ yields the unique analytical closed-form solution used by our Python script:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad (7)$$